

Chetan Joshi

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Education

- **University of Utah** SLC, UT
PhD in Computer Science May 2024 - Present
Graduate Research Assistant
- **Brigham Young University** Provo, UT
MS in Computer Science Sep 2021 - April 2024
Research Assistant, Full stack SWE
- **G. B. Pant University of Agriculture & Technology** Pantnagar, India
Bachelor of Computer Engineering Aug 2014 - June 2018

Publications

- **Head Tail Open: Open tailed classification of imbalanced document data.** (SAI'24)
- **CENSUS-HWR: a large training dataset for offline handwriting recognition.** (CSCI'23)

Skills Summary

Python, Pytorch, Triton, Cuda, TVM

Research Experience

Graduate Research Assistant, UOU May 2024 - Present
High Performance Computing Lab SLC, UT

- Characterized the search space for autotuning CNN kernels based on their geometry and family, which helped in pruning the search space by 20% for standard convolutions.
- Improved the speed of a ResNet-50 model by 30% by vectorizing sequential code for Global Average Pooling operator and maintained the code correctness.
- Reduced MobileNet V2 model size by 1/6th using Neural Architecture Search (NAS) and a proxy score, while maintaining 90% accuracy on an animal ecology datasets.

Graduate Research Assistant, BYU Sep 2021 - April 2023
Family History Technology Lab Provo, UT

- Improved an open set document classification model by leveraging the embedding capabilities of the Segment Anything Model, achieving an accuracy of 81%.
- Developed deep learning models for handwriting recognition, document understanding, NER and text classification tasks using Tensorflow, and achieved 85% accuracy and sorted more than 200,000 historical records.
- Enhanced accuracy by 25% by optimizing the model, applying data augmentation, & handling imbalanced data.

Professional Experience

Data Science Intern May 2022 - Aug 2022
Ancestry Lehi, UT

- Studied the deep learning (transformer architecture) based recommendation system which recommends personalized items and optimized it to increase the diversity of the recommended items by 20%.
- Engineered 18 new feature columns that aimed to diversify the personalized recommended items and increase the customer retention rate by 17%.
- Conducted data analysis & visualization, presented useful insights to the data science team.

Full Stack SWE Sep 2018 - Aug 2021
BYU Broadcasting Provo, UT

- Built backend services for event snapshotting in an event-driven, domain-driven architecture.
- Resolved platform issues across Web, Xbox, and Samsung clients using JavaScript, Node.js, and Vue 2.
- Developed and debugged applications while collaborating in an agile engineering team.

Software Engineer
Tata Consultancy Services

Sep 2018 - Aug 2021
Indore, India

- Implemented over 20 REST APIs in Java using the MVC design and object oriented programming, optimizing manual tasks by 40%.
- Ensured code maintainability, usability and scalability, conducted code reviews, mentored new hires, and introduced them to our SDLC process.
- Analyzed the complex data using SQL, developed Junit tests, located error sources and updated the code, resulting in 80% reduction of defects.

Projects

Optimizing Vision Transformers with Triton

Jan 2025 - April 2025

- Implemented a custom FlashAttention 2.0 kernel in Triton for ViT, eliminating large intermediate matrices via block-wise GPU SRAM-aware computation.
- Reduced ViT inference latency by up to 13.2% compared to the Hugging Face baseline while preserving numerical accuracy and model performance.

Warpscope

Feb 2026 - April 2026

- Developed an NVBit-based GPU profiler that instruments CUDA kernels to detect and quantify warp idleness using barrier and compute fraction metrics (BarrFrac/CompFrac)
- Applied warp idleness insights to guide warp specialization in FlashAttention kernels, achieving a $3.7\times$ reduction in warp idle time on KV-cache decode workloads